newlength (E) = Dist (D) + w(D,E) = 220 + 20 = 240
newlength (F) = Dist (D) + w(D,F) = 220 + 40 = 260

Iteration 5
S ∈ (B, J)
newlength (B) = Dist (S) + w(S,B) = 620 + 500 = 1150
newlength (J) = Dist (S) + w(S,J) = 620 + 480 = 1100

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Question 5

Test table

#	source, destination	expected	output	result
1	A, T	shortest route ~~through~~ through network	A → Q → T distance 310	pass
2	A, A (same node)	Distance 0	path 'A', distance 0	pass
3	S, E (far apart	Long multi-hop route	S → B → C → D → E, distance 620	pass
4	c, p (lowercase)	Treated as C, P	C → N → L → M → P distance 220	pass
5	B, S (direct road)	Direct edge is shortest	B → S, distance 500	pass
6	A, S	Goes via B	A → B → S distance 620	pass
7	Z, B (invalid letter)	Error message	Error message, Program exits with code 1	pass
8	C, 5 (invalid character	Error message	Error message, exit code 1	pass